Prayas JEE AIR (2025)

WEEKLY TEST 29

DURATION : 180 Minutes | **DATE : 11/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

Single Correct Type Questions

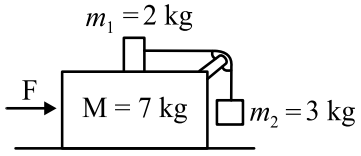
1. The relation between time t and distance x for a moving body is given as $t = 3x^2 + 2x$. The retardation of the moving body at $x = \frac{1}{3}$ m is?

(1) $\frac{5}{32}$ m/sec^2 (2) $\frac{3}{64}$ m/sec^2

(3) $\frac{3}{32}$ m/sec^2 (4) $\frac{5}{64}$ m/sec^2

2. Three masses M = 7 kg, m_1 = 2 kg and m_2 = 3 kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F = 236 N is applied on the system so that the mass m_2 moves upward with an acceleration of a ms^{-2}. The value of a is

(Take g = 10 ms^{-2})



(1) 2 (2) 4

(3) 6 (4) 10

3. An automobile of mass m = 90 kg accelerates starting from origin and initially at rest, while the engine supplies constant power P = 60 W. The position of automobile at time t = 3 sec. is.

(1) x = 4 m (2) x = 6 m

(3) x = 8 m (4) x = 9 m

4. A tennis ball is released from a height of 8.1 m above the floor. It bounces back repeatedly, always rising to $\frac{64}{100}$ of the height through which it falls. Find the approximate average speed of the ball by the time it completely come to rest.

(Take g =10 ms^{-2})

(1) 3.2 m/sec.

(2) 2 m/sec.

(3) 6.2 m/sec.

(4) 5 m/sec.

5. Moment of inertia of a cylinder of mass M, length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right)$. If such a cylinder is to be made for a given mass of a material, then the minimum possible value of its radius of gyration about such axis is

(1) $\frac{L}{3}$ (2) $L\sqrt{\frac{3}{2}}$

(3) $\frac{L}{2}$ (4) $L\sqrt{\frac{2}{3}}$

6. The value of the acceleration due to gravity is g_1 at a depth $d = \frac{R}{2}$ (R = radius of the earth) from the surface of the earth. It is again equal to g_1 at a height h above the surface of the earth. The ratio $\left(\frac{h}{R}\right)$ equals

(1) 0.81 (2) 0.51

(3) 0.41 (4) 0.61

7. Four identical symmetrically placed hollow cylindrical columns of steel support a big structure of mass 50 × 10^3 kg. The inner and outer radii of each column are 50 cm and 100 cm respectively. Assuming uniform local distribution, calculate the strain energy density (in mJ/m^3) of each column.

[Use Y = 2.0 ×10^{11} Pa, g = 10 m/s^2]

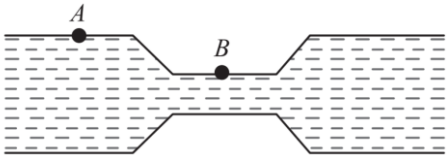
(1) 6.94

(2) 2.95

(3) 9.96

(4) 5.84

8. Water flows in a horizontal tube (see figure). The pressure of water changes by 375 N/m^2 between A and B where the area of cross section are 60.4 cm^2 and 30.2 cm^2, respectively. Find the rate of flow of water through the tube. (density of water =1000 kg m^{-3})



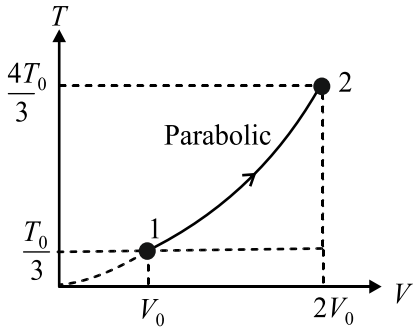
(1) 1810 cm^3/s

(2) 2420 cm^3/s

(3) 2720 cm^3/s

(4) 3020 cm^3/s

9. If the process on one mole of monatomic ideal gas is as shown in the *TV*-diagram



	Column-I		Column-II
A	Work done by the system in the process / RT_0	P	$\frac{1}{2}$
B	Change in internal energy in the process / RT_0	Q	$\frac{1}{3}$
C	Heat absorbed by the system in the process / RT_0	R	$\frac{3}{2}$
D	Specific heat capacity in the process / R	S	$\frac{2}{3}$
		T	2

The correct match is

(1) A → S; B → T; C → Q; D → P

(2) A → P; B → R; C → T; D → S

(3) A → P; B → S; C → Q; D → T

(4) A → P; B → R; C → T; D → T

10. A simple pendulum, made of a string of length l and a bob of mass m = 1 kg, is released from a small angle θ_0. It strikes a block of mass M = 2 kg, kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1. Then $\frac{\theta_1}{\theta_0}$ is.

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

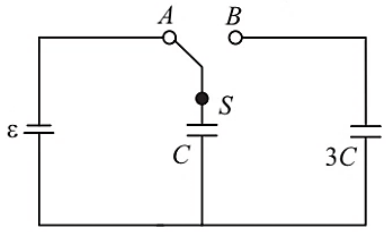
(3) $\frac{1}{4}$ (4) $\frac{1}{5}$

11. In a cuboid of dimension $L \times L \times \frac{L}{2}$, a charge $2q$ is placed at the centre of the surface '*S*' having area of L^2. The flux through the opposite surface to '*S*' is ϕ_1, and the flux through any one adjacent surface is ϕ_2, than find the value of $\phi_1 + \phi_2$

(1) $\frac{q}{2\varepsilon_0}$ (2) $\frac{q}{3\varepsilon_0}$

(3) $\frac{q}{4\varepsilon_0}$ (4) $\frac{q}{6\varepsilon_0}$

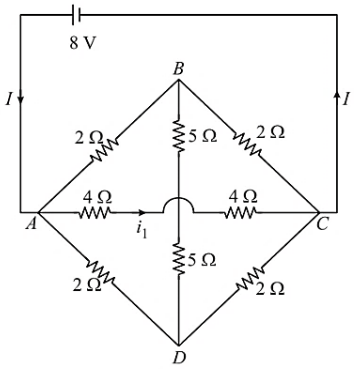
12. In the figure shown, after the switch '*S*' is turned from position '*A*' to position '*B*', the fraction of energy stored on capacitor dissipated in the circuit after closing the switch, is



(1) $\frac{3}{8}$ (2) $\frac{1}{8}$

(3) $\frac{5}{8}$ (4) $\frac{3}{4}$

13. Find the rate at which energy is delivered by the battery.



(1) 10 W (2) 20 W

(3) 30 W (4) 40 W

14. A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic pressure in the wire at distance $\frac{a}{3}$ and $\frac{a}{2}$, from the axis of the wire respectively is

(1) $\frac{16}{9}$

(2) $\frac{32}{27}$

(3) $\frac{9}{16}$

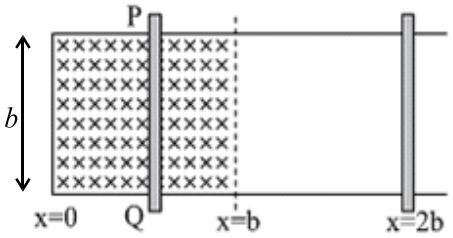
(4) $\frac{27}{32}$

15. The arm PQ of a rectangular conductor is moving from $x = 0$ to $x = 2b$ has resistance R as shown in the figure. A magnetic field perpendicular to the plane is acting from $x = 0$ to $x = b$ is given by

$$B(x) = -B_0\left(1-\frac{x^2}{b^2}\right)\hat{k}$$

and $B = 0$ from $x = b$ to $x = 2b$.

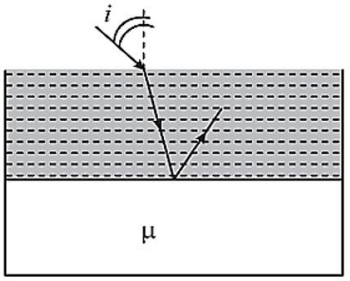
The amount of charge flown through the conductor is. (Negater the resistance of remaining arms)



(1) $\frac{B_0b^2}{R}$ (2) $\frac{2B_0b^2}{3R}$

(3) $\frac{3B_0b^2}{2R}$ (4) $\frac{B_0b^2}{2R}$

16. Consider a tank with a thick bottom of refractive index μ. It is filled with a liquid of refractive index 4/3. A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid solid interface is never completely polarized. For this to happen, the minimum value of μ is



(1) $\sqrt{\frac{5}{3}}$ (2) $\frac{4}{\sqrt{7}}$

(3) $\frac{5}{\sqrt{3}}$ (4) $\frac{4}{3}$

17. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): The phase difference of two light waves change if they travel through different media having same thickness, but different indices of refraction.

Reason (R): The wavelengths of waves are different in different media.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both A and R are correct but R is not the correct explanation of A

(2) A is correct but R is not correct

(3) Both A and R are correct and R is the correct explanation of A

(4) A is not correct but R is correct

18. A nucleus X, with a finite de-broglie wavelength λ_0 undergoes spontaneous fission into two nuclei Y and Z of equal mass. Y flies in the same direction as that of X, while Z flies in the opposite direction with a velocity equal to one third of Y. The de-Broglie wavelengths of Y and Z are respectively λ_Y and λ_Z . Then find the value of $\frac{\lambda_Y+\lambda_Z}{\lambda_0}$

(1) $\frac{8}{3}$ (2) $\frac{1}{3}$

(3) $\frac{3}{8}$ (4) $\frac{1}{8}$

19. Which level of the single ionized helium ion has the same energy as the double ionized lithium ion in its 5th excited state?

(1) 6 (2) 8

(3) 4 (4) 1

20. Find the approximate value of binding energy per nucleon for ${}_{26}Fe^{56}$. Mass of proton m_P = 1.00783 u, mass of neutron m_n = 1.00867 u and mass of iron nucleus $m_{F_e} = 55.845\,\text{u}$

(Take 1 u = 931 MeV/c^2)

(1) 9.0 MeV (2) 10.3 MeV

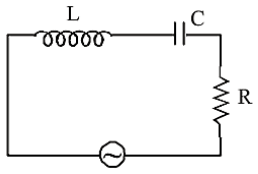
(3) 8.0 MeV (4) 11.5 MeV

Integer Type Questions

21. A wheel rotates with a constant acceleration of 2.0 radian/sec^2. If the wheel starts from rest, then by the end of first ten seconds the wheel is near to completion of nth revolution. The value of n will be

22. Two concentric coplanar circular loops made of wire, with resistance per unit length 0.1 Ω/m have diameters 0.2 m and 2 m. A time varying potential difference 4 (1 + t) volt is applied to the larger loop. Calculate the current (in μA) in the smaller loop close to nearest integer.

23. Given LCR circuit has L=5H, C = 80 μF, R = 40 Ω and variable frequency source of 200V. The source frequency (in Hz) which drives the circuit at resonance is $\frac{x}{\pi}$. Find the value of x.



24. Light of wavelength 155 nm ejects photoelectron from a plate of a metal whose work function is 2eV. If a uniform magnetic field of $B = \sqrt{\frac{5}{6}} \times 10^{-3}$T is applied parallel to plate, what would be the radius (in mm) of the path followed by electrons ejected normally from the plate with maximum energy?
(given hc = 1240 eV–nm & consider $m_e = 9 \times 10^{-31}$ kg)

25. At 40° C, a brass wire of radius $\frac{1}{\sqrt{\pi}}$ mm is hung from the ceiling. *A* small mass, *M* is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of *M* (in kg) is close to: (Coefficient of linear expansion and Young's modulus of brass are 10^{-5}/°C and 10^{11} N/m^2, respectively; g =10 ms^{-2}).

SECTION-II (CHEMISTRY)

Single Correct Type Questions

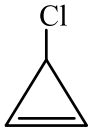
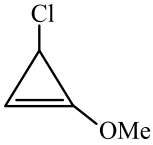
26. Propyne and propene can be distinguished by
(1) conc. H_2SO_4
(2) Br_2 in CCl_4
(3) $AgNO_3$ [Ammoniacal]
(4) dil. $KMnO_4$

27. Which one of the following order represents the correct sequence of the increasing basic nature of the given oxides?
(1) $MgO < K_2O < Al_2O_3 < Na_2O$
(2) $Na_2O < K_2O < MgO < Al_2O_3$
(3) $K_2O < Na_2O < Al_2O_3 < MgO$
(4) $Al_2O_3 < MgO < Na_2O < K_2O$

28. For an elementary chemical reaction, $A_3 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 3A$, the expression for $\left(-\frac{d[A_3]}{dt}\right)$ is
(1) $k_1[A_3] + k_{-1}[A]^3$
(2) $3k_1[A_3] - 3k_{-1}[A]^3$
(3) $3k_1[A_3] - k_{-1}[A]^3$
(4) $k_1[A_3] - k_{-1}[A]^3$

29. The dipole moment is lowest for which of the following compound?
(1) $CHCl_3$
(2) CCl_4
(3) CH_2Cl_2
(4) CH_3Cl

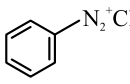
30. Cellulose is composed of:
(1) α-D-glucose, C_1–C_4 linkages
(2) α-D-glucose, C_2–C_6 linkages
(3) β-D-glucose, C_2–C_6 linkages
(4) β-D-Glucose, C_1–C_4 linkages

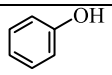
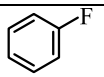
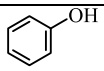
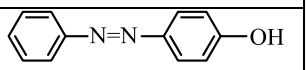
31. The most reactive molecule among the following organic molecules towards $AgNO_3$ solution is
(1) 
(2) 
(3) $CH_3CH(Cl)CH_3$
(4) $CH_3CH(Cl)CH_2NO_2$

32. The relative stability of +3 oxidation state of group 13 elements follows the order:
(1) Al < Ga < Tl < In
(2) Tl < ln < Ga < Al
(3) Ga < Al < ln < Tl
(4) Al < Ga < ln < Tl

33. Choose the correct set of reagents for the following conversion trans (CH_3-CH=CH–CH_3) → cis (CH_3-CH=CH– CH_3)
(1) Cl_2, alc KOH, $NaNH_2$, Na(Liq NH_3)
(2) Cl_2, alc KOH, $NaNH_2$, H_2, Lindlar Catalyst
(3) Cl_2, aq KOH, $NaNH_2$, H_2, Lindlar Catalyst
(4) Cl_2, aq KOH, $NaNH_2$, Na(Liq NH_3)

34. If the de-Broglie wavelength of the electron in nth Bohr orbit in a hydrogenic atom is equal to 2.5 πa_0 (a_0 is Bohr radius), then the value of z/n is:
(1) 0.80 (2) 1.50
(3) 1.0 (4) 0.75

35. Match List-I with List-II  is reacted with reagents in List-I to form products in List-II

	Column-I (Reagent)		**Column-II (Product)**
A		P	
B	HBF_4, Δ	Q	
C	H_3PO_2	R	
D	H_2O,H_2SO_4, 100°C	S	

Choose the correct answer from the options given below:

(1) A→S; B→R; C→Q; D→P
(2) A→R; B→P; C→S; D→Q
(3) A→R; B→P; C→Q; D→S
(4) A→P; B→R; C→S; D→Q

36. For the reaction $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g)$ the pressure of $CO_2(g)$ depends on

(1) Amount of $SrCO_3(s)$
(2) Amount of SrO (s)
(3) Amounts of both $SrCO_3(s)$ and SrO(s)
(4) Temperature of the system

37. In the following reaction

$$\text{Aldehyde} + \text{Alcohol} \xrightarrow{HCl} \text{Acetal}$$

HCHO t – BuOH

Ph – CHO MeOH

The best combination is

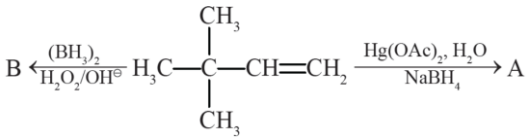
(1) HCHO and t-BuOH
(2) Ph–CHO and t-BuOH
(3) HCHO and MeOH
(4) Ph–CHO and MeOH

38. The difference between ΔH and ΔU (ΔH – ΔU), when the combustion of one mole of octane (*l*) is carried out at a temperature T, is equal to consider H_2O is evolved in liquid state at temperature. T & octane is also in liquid state

(1) 3RT (2) $-\frac{9}{2}RT$

(3) –4RT (4) $\frac{5}{2}RT$

39. Choose the incorrect option for the following reactions



(1) 'A' and 'B' are both addition products
(2) 'A' is Markovnikov product
(3) 'B' is anti-Markovnikov product.
(4) 'B' is Markovnikov product.

40. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The energy required to form Mg^{2+} from Mg is less than that required to produce Mg^{+}.

Reason (R): Mg^{2+} is small ion and carries more charge than Mg^{+}.

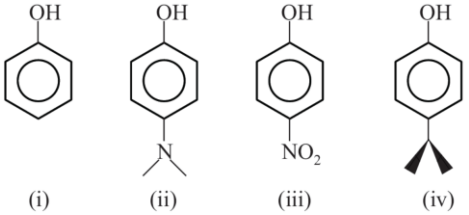
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(2) (A) is true but (R) is false.
(3) (A) is false but (R) is true.
(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

41. The crystal field stabilization energy (CFSE) of $[Ti(H_2O)_6]^{3+}$ is:

(1) $-0.8\Delta_0$
(2) $-0.6\ \Delta_0$
(3) $-0.4\Delta_0$
(4) $-1.2\ \Delta_0$

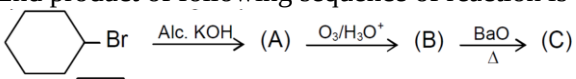
42. The correct order of acidic strength for the following compounds is:

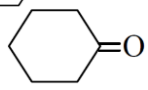
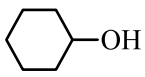
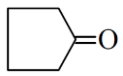
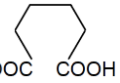


(i) (ii) (iii) (iv)

(1) (iii) > (i) > (iv) > (ii)
(2) (ii) > (iv) > (i) > (iii)
(3) (ii) > (i) > (iv) > (iii)
(4) (i) > (ii) > (iii), (iv)

43. End product of following sequence of reaction is



(1) 
(2) 
(3) 
(4) 

44. Molal depression constant for a solvent is 3.0 K Kg mol^{-1}. The depression in the freezing point of the solvent for 0.05 mol kg^{-1} solution of $BaCl_2$ is: (Assume complete dissociation of the electrolyte)

(1) 0.12 K
(2) 0.36 K
(3) 0.18 K
(4) 0.45 K

45. 2 moles of SO_2 and 1 mole of O_2 are allowed to react in a container of capacity 2 litre and following equilibrium is allowed to establish:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

The equilibrium mixture requires 200 ml of 0.1 M acidified $K_2Cr_2O_7$ solution. Hence Kc for the above equilibrium is approximately

(1) 6.96×10^2 (2) 6.96×10^3
(3) 6.96×10^4 (4) 6.96×10^5

Integer Type Questions

46. 200 g of propane is completely reacted with 2000 g of oxygen. The mole fraction of water molecules formed in the resulting mixture is x $\times 10^{-2}$. The value of x is _____.
(Nearest integer)
[Atomic weight: H=1.008; C=12.00; O = 16.00]

47. A 6.0 m mol dm^{-3} aqueous solution of KCl has a conductance of 0.66 mS when measured in a cell of cell constant 1.2 cm^{-1}. The molar conductivity of this solution is_____ S cm^2 mol^{-1}.
(Round off to the nearest integer).

48. The value of azimuthal quantum number of the outermost electron of Zn^+ ion is _____.

49. (S) – 2 – chloropentane on reaction with Cl_2 (g) in the presence of U.V. light, gives 'N' number of optically active dichloro derivatives. Hence the value of N is

50. 400 mL of 0.5 M NaOH was added to 600 mL of 1M HCl. The number of unreacted HCl molecules in the solution after complete reaction is_____ $\times 10^{21}$ (Nearest integer)
($N_A = 6.022 \times 10^{23}$)

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let *S* be set of all real roots of the equation, $x^2 - x + 2 = |x - 1| + |x - 2|$. Then *S*

(1) contains exactly two elements
(2) is a singleton
(3) contains at least four elements
(4) is an empty set

52. Matrices of order 3×3 are formed by using the elements of the set $A = \{-4, -2, 0, 2, 4\}$ then the number of either symmetric or skew-symmetric matrices is-

(1) $5^6 + 5^3 - 1$
(2) $5^5 + 3^6 + 1$
(3) $6^5 + 3^5 - 1$
(4) $5^3 + 5^6 + 1$

53. The sum of all the coefficients of x in the product $(1 + x)\ (1 - x)^{10}\ (1 + x + x^2)^9$ is:

(1) –84 (2) 0
(3) 126 (4) 84

54. Let arg(*z*) represent the principal argument of the complex number *z*. Then $\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$ is represented by the equation of circle/circular arc having center at

(1) (0, 0) (2) (1, 0)
(3) (0, 1) (4) (1, 1)

55. The equations of the sides *AB*, *BC* and *CA* of a triangle *ABC* are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is $\left(2, a\right), -\frac{1}{2} < a < 2$, then $p - a + 1$ is equal to

(1) 1
(2) 2
(3) 3
(4) 4

56. If the shortest between the lines

$\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and

$\frac{x+2\sqrt{6}}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is

(1) 2
(2) 3
(3) 6
(4) 9

57. Let the mean of 6 observation 1, 2, 4, 5, *x* and *y* be 5 and their variance be 10. Then *xy* is equal to

(1) 20
(2) 40
(3) 60
(4) 80

58. Let N be the sum of the number appeared when two fair dice are rolled. The probability that $N-2, \sqrt{3N}, N+2$ are in geometric progression is

(1) $\frac{1}{24}$ (2) $\frac{1}{12}$

(3) $\frac{1}{3}$ (4) $\frac{1}{6}$

59. If the area of the triangle whose one vertex is the vertex of the parabola, $y^2 + 4(x - 25) = 0$ and the other two vertices are the points of intersection of the parabola and y – axis, is

(1) 125 (2) 500

(3) 375 (4) 250

60. Let $f: [0, \infty) \to [0, 3]$ be a function defined by

$$f(x) = \begin{cases} \sin x, & 0 \le x \le \frac{\pi}{2} \\ 1, & \frac{\pi}{2} < x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$$ Then which of the following is true?

(1) f is continuous and differentiable everywhere in $(0, \infty)$

(2) f is not continuous exactly at two points in $(0, \infty)$

(3) f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$

(4) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

61. The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1, 3, 5, 7, 9, with repetition of digits allowed, is

(1) 25 (2) 125

(3) 275 (4) 375

62. The number of possible positive values of x for which $\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$ is

(1) zero (2) 1

(3) 2 (4) infinitely many

63. Let $R_1 = \{(a,b) \in N \times N : |a-b| \le 11\}$ and $R_2 = \{(a,b) \in N \times N : |a-b| \ne 21\}$. Then on N

(1) Both R_1 and R_2 are equivalence relations

(2) Neither R_1 nor R_2 is an equivalence relations

(3) R_1 is an reflexive relation but R_2 is not

(4) R_2 is an symmetric relation but R_1 is not

64. If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b)$ meets the line $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the length of major axis of the ellipse is

(1) 14 (2) $4\sqrt{6}$

(3) 7 (4) $2\sqrt{6}$

65. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$ be three vectors. If $\vec{r}$ is a vector such that, $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $5\vec{r} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$. The value of $\alpha + \beta + \gamma$ is

(1) 4 (2) 6

(3) 9 (4) 0

66. Let the hyperbola $3x^2 - 4y^2 = 36$. The eccentricity of it's conjugate hyperbola is

(1) $\frac{\sqrt{7}}{2}$ (2) $\sqrt{7}$

(3) $\sqrt{\frac{7}{3}}$ (4) $\frac{\sqrt{7}}{4}$

67. If a curve $y = f(x)$ passing through $(4, -2)$ and satisfying the differential equation $(y^3 - x)\,xdy = (x + y^3)\,ydx$ then equation of curve is

(1) $2x + y^2 = 0$ (2) $2x + y = 0$

(3) $2x + y^3 = 0$ (4) $2x^2 + y = 0$

68. The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0, x = \frac{\pi}{4}$ is

(1) $2 - \sqrt{2}$

(2) $(\sqrt{2} + 1)$

(3) $4(\sqrt{2} - 1)$

(4) $2\sqrt{2}(\sqrt{2} + 1)$

69. Let 'a' be a real number such that the function $f(x) = ax^2 - 6x + 15$, $x \in R$ has a local extremum $x = -3/4$. Then the function $g(x) = ax^2 + 6x - 15$, $x \in R$ is increasing in.

(1) $x \in R$

(2) $x \in \phi$

(3) $x \in \left(\frac{3}{4}, \infty\right)$

(4) $x \in \left(-\infty, \frac{3}{4}\right)$

70. If the number of integral terms in the expansion of $\left(3^{1/2}+5^{1/8}\right)^{256}$ is

(1) 257
(2) 128
(3) 33
(4) 223

Integer Type Questions

71. The value of $\left[\frac{10}{\pi}\int_0^{\sqrt{2}}\frac{\left(2-x^2\right)dx}{\left(2+x^2\right)\sqrt{4+x^4}}\right]$ is equal to

[where [.] represents the greater integer function]

72. The radius of the circle that is the locus of a point which moves such that the sum of squares of its distances from the points (0, 0), (1,0), (0,1), (1,1) is 18 units, is equal to

73. If $\int\frac{\sin x}{\sin^3 x+\cos^3 x}dx$

$=\log_e|1+\tan x|^{\alpha}+\log_e\left|1-\tan x+\tan^2 x\right|^{\beta}$

$+\gamma\tan^{-1}\left(\frac{2\tan x-1}{\sqrt{3}}\right)+C$, when C is constant of integration then the value of $\frac{1}{\alpha+\beta+\gamma^2}$ is ____

74. A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is a spade, is p/q (where p & q are coprime) then $|q-p|$ is

75. If $\sin\left(10^\circ\right)\sin\left(20^\circ\right)\sin\left(40^\circ\right)\sin\left(50^\circ\right)\sin\left(70^\circ\right)$ $\sin(80°)=\frac{\sqrt{3}}{\alpha}$ then α is equal to



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